

Ex. No. 9	ESTABLISHING CONNECTION BETWEEN MONGODB (COMPASS) AND JAVA(ECLIPSE)
Date: 16-03-2026	

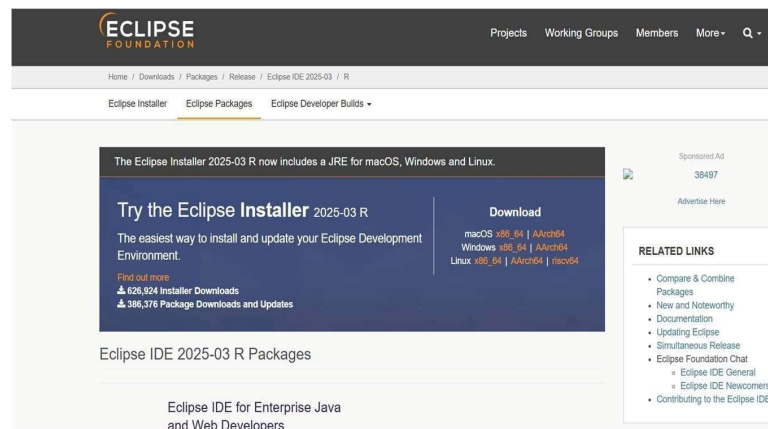
AIM:

To establish a connection between a Java program and a MongoDB database using Maven dependencies, insert a document into the database, and verify the insertion through MongoDB Compass.

PROCEDURE:

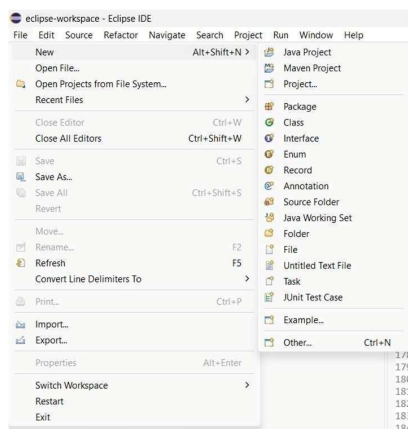
Step 1: Set Up the Environment

- Install **Eclipse IDE** from <https://eclipseide.org/> (if not already installed).



Step 2: Create a New Java Project

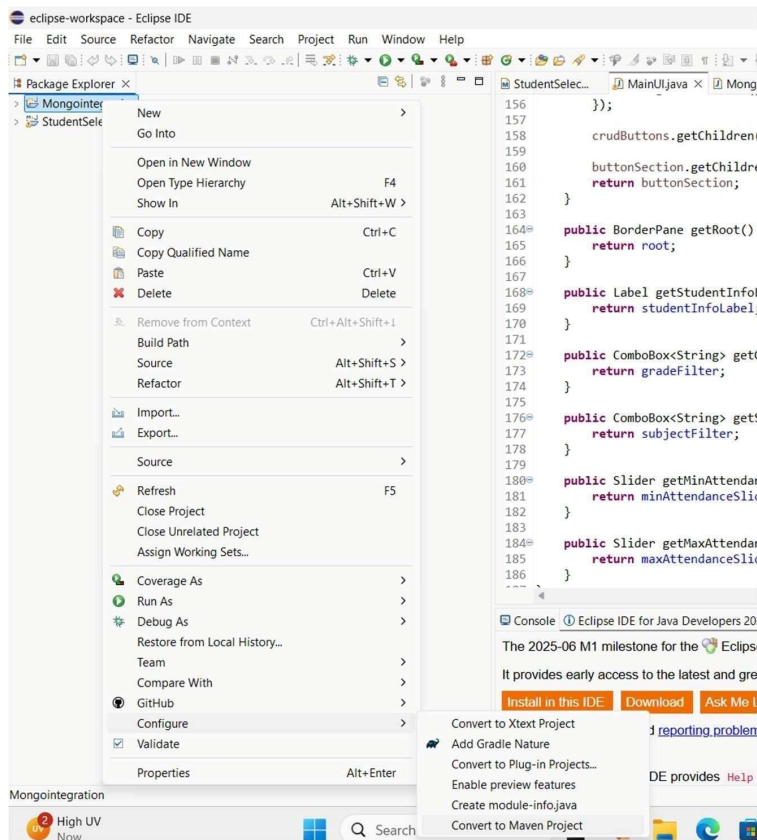
- Open Eclipse.
- Go to File > New > Java Project.



- Name the project (e.g., MongoIntegration).

Step 3: Convert to Maven Project

- Right-click the project > Configure > Convert to Maven Project.



Step 4: Add Maven Dependency

- Open the pom.xml file.
- Add the following dependency inside the <dependencies> tag:

```
<dependency>
<groupId>org.mongodb</groupId>
<artifactId>mongodb-driver-sync</artifactId>
<version>4.11.0</version> </dependency>
```

```

https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1=<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://m
2  <modelVersion>4.0.0</modelVersion>
3  <groupId>MongoIntegration</groupId>
4  <artifactId>MongoIntegration</artifactId>
5  <version>0.0.1-SNAPSHOT</version>
6  <build>
7    <sourceDirectory>src</sourceDirectory>
8  <plugins>
9    <plugin>
10     <artifactId>maven-compiler-plugin</artifactId>
11     <version>3.13.0</version>
12     <configuration>
13       <release>21</release>
14     </configuration>
15   </plugin>
16 </plugins>
17 </build>
18 </project>

```

- Save the pom.xml. Maven will automatically download the MongoDB driver library.

Step 5: Write Java Code

- Create a Java class (e.g., MongoConnect.java).
- Write the following code to connect to MongoDB, insert data, and retrieve documents:

```

import
com.mongodb.client.*; import
org.bson.Document;

public class MongoConnect { public
    static void main(String[] args)
    { String uri =
      "mongodb://localhost:27017";
      MongoClient mongoClient = MongoClient.create(uri);

      MongoDBDatabase database = mongoClient.getDatabase("testDB");
      MongoCollection<Document> collection = database.getCollection("students");

      Document doc = new Document("name", "Sahana")
        .append("age", 21)
        .append("department", "IT");
      collection.insertOne(doc);

      FindIterable<Document> docs =
      collection.find(); for (Document d : docs) { System.out.println(d.toJson());
      }

      mongoClient.close();
    }
}

```

Step 6: Run the Program

- Right-click the file > Run As > Java Application.
- Observe the console output displaying inserted documents.

Step 7: Verify in MongoDB Compass

- Open MongoDB Compass.
- Connect to mongodb://localhost:27017.
- Navigate to **testDB** > **students** collection.
- Verify that the document (e.g., Sahana, 21, IT) is inserted.

Result:

Thus, the connection between MongoDB (compass) and JAVA(eclipse) is established successfully.

